JUAN DAVID BENDEK WILLIAMSON

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Profile

Mechanical engineer and MSc graduate specialising in data science, computational modelling, robotics and agentic AI. Experience spans enterprise MCP/SQL-agent development, ROS 2 cognitive robotics, structural simulation and bilingual technical support.

Core Skills

Programming and Data
Python, SQL, MATLAB, Git, data analysis, reporting, workflow design and automation.

AI and ML
LLM agents, MCP, SQL agents, tool calling, LiteLLM, llama.cpp, reinforcement learning, PPO, GAE and LoRA/QLoRA concepts.

Robotics and Modelling
ROS 2, NAO, HRI interfaces, LLM task planning, Abaqus, ANSYS Fluent, Mathcad, STAAD.Pro, Inventor and SolidWorks.

Languages
Fluent English and Spanish; A1 Hungarian.

Selected Research

MSc Thesis - NAO LLM Stack
Designed and validated a modular ROS 2/Jazzy NAO stack separating dialogue, symbolic grounding, LLM planning, deterministic execution, feedback-driven replanning and truthful reporting.

BSc Thesis
Built and validated a neural-network model for turbulent-intensity prediction from wind-profile data.

Education

MSc in Modelling for Science and Engineering — Sep 2025 - Jul 2026
Universitat Autonoma de Barcelona
Data Science specialisation. Thesis: modular language-model-enabled interaction and execution stack for the NAO robot.

BSc in Mechanical Engineering — Sep 2020 - Jul 2024
Budapest University of Technology and Economics
Stipendium Hungaricum scholar; Process Engineering specialisation; final grade 4.33 GPA.

A-levels: Physics, Mathematics and Economics — Sep 2017 - Jun 2019
Farnborough Sixth Form College
Mathematics: A; Economics: B; Physics: B; UKMT Maths Challenge Bronze.

Experience

Aily Labs — 2026 - Aug 2026
Intern Data Scientist
- Developed MCP-enabled skills, SQL-agent interfaces and reusable AI/data capabilities for enterprise analytics and GTM decision support.
- Translated analytical requirements into structured context, semantic interfaces, DAG-style workflows, reporting components and testable agent outputs.
- Supported migration, documentation and validation while preserving clear boundaries between data, tools, prompts and evaluation logic.

IIIA-CSIC — Jan 2026 - Aug 2026
JAE Intro ICU 2025 Scholar - LLM-based Architectures for Interactive Systems
- Researched modular LLM-based cognitive robotics architectures for interactive human-robot systems and domestic-like tasks.
- Implemented ROS 2/NAO pipelines connecting dialogue, intent, grounded symbolic state, structured planning, deterministic execution and feedback.

ABS Consulting — Nov 2024 - Oct 2025
Engineer I
- Supported seismic qualification and structural assessment projects for UK nuclear power plant assets.
- Built and reviewed FEA and engineering models using Abaqus, Mathcad, Inventor and STAAD.Pro; prepared billable technical reports.
- Applied Eurocodes and British Standards to calculations and client deliverables.

Transcom Ltd — Aug 2021 - Jul 2024
Technical Support Center Level 1 - UPS Project
- Delivered Level 1 technical support in English and Spanish through calls, email and Microsoft Teams.
- Supported UPS employee, warehouse and shipment-flow systems; escalated issues through clear SMC tickets.

Technical Projects

- NAO LLM stack (MSc thesis): built a modular ROS 2/Jazzy architecture for dialogue, KB grounding, structured planning, deterministic skill execution, feedback-driven replanning and scenario-based validation.
- iTrader: developed an RL trading research framework with market simulation, proposer/solver loops, task specifications, reward registries and PPO+GAE experimentation.
- Watson Stack: integrated Hermes, LiteLLM and llama.cpp for local coding, research, file-tooling and model-routing workflows.